

MSTAR PRACTICE WORKSHEET**Lesson 8-4: Write and Represent Inequalities**

Grade 6 Mathematics · Standard 6.AT.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'The temperature is less than 40 degrees'?

- ☒ A $t < 40$
- ☐ B $t > 40$
- ☐ C $t \leq 40$
- ☐ D $t = 40$

'Less than' means $<$, so the inequality is $t < 40$.

Question 2 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'You must be at least 48 inches tall to ride'?

- ☒ A $h \geq 48$
- ☐ B $h > 48$
- ☐ C $h < 48$
- ☐ D $h \leq 48$

'At least' means greater than or equal to, so $h \geq 48$.

Question 3 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'A store can hold no more than 75 people'?

- ☒ A $p \leq 75$
- ☐ B $p < 75$
- ☐ C $p \geq 75$
- ☐ D $p > 75$

'No more than' means less than or equal to, so $p \leq 75$.

Question 4 SELECTED RESPONSE — CORRECT: A

Which inequality represents: 'The speed limit is more than 25 mph'?

- ☒ A $s > 25$
- ☐ B $s \geq 25$
- ☐ C $s < 25$
- ☐ D $s = 25$

'More than' means strictly greater than, so $s > 25$.

Question 5 SELECTED RESPONSE — CORRECT: A

A parking garage charges if your stay is more than 2 hours. Which inequality represents times t that are charged?

- ☒ A $t > 2$
- ☐ B $t \geq 2$
- ☐ C $t < 2$
- ☐ D $t \leq 2$

'More than 2 hours' means strictly greater than: $t > 2$. Exactly 2 hours is not charged.

Question 6

SELECTED RESPONSE — CORRECT: A

A clue says the suspect carried at least 12 stolen gems. Which inequality shows the number g of gems?

A $g \geq 12$

B $g > 12$

C $g \leq 12$

D $g < 12$

'At least 12' includes 12 and any number above it, so $g \geq 12$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: A**

A fair ride requires each rider to be at least 48 inches tall. Let h be a rider's height in inches. Which inequality represents the heights that are allowed to ride?

- ☒ **A** $h \geq 48$
- ☐ **B** $h > 48$
- ☐ **C** $h \leq 48$
- ☐ **D** $h < 48$

The phrase 'at least 48 inches' means 48 inches is allowed and any greater height is allowed too. That is greater than or equal to, so the inequality is $h \geq 48$.

- **B:** This shuts out a rider who is exactly 48 inches tall. 'At least' includes the 48.
- **C:** This admits only riders 48 inches and SHORTER. Taller riders are allowed too.
- **D:** This admits only riders under 48 inches, which is the opposite of the rule.

Part B — correct answer: C

Which explanation justifies the symbol chosen above?

- ☐ **A** The words 'at least' mean a rider must be taller than 48 inches, so 48 inches is not allowed.
- ☐ **B** The words 'at least' point to the smallest numbers, so riders must be shorter than 48 inches.
- ☒ **C** The words 'at least' include the boundary value, so a rider who is exactly 48 inches tall may ride.
- ☐ **D** The words 'at least' and 'at most' describe the same group of heights, so either symbol works.

'At least' sets a minimum that is part of the group, which is why the symbol carries the bar underneath. A rider who measures exactly 48 inches meets the rule, so $h \geq 48$ rather than $h > 48$.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Reversing 'No More Than'

Priya read this situation: 'A backpack can hold no more than 15 pounds.' She let w stand for the number of pounds in the backpack and wrote the inequality $w \geq 15$.

Explain Priya's misconception and give the correct inequality.

Model answer: $w \leq 15$. 'No more than 15 pounds' means the weight may be 15 pounds or anything below 15 pounds, so the correct inequality is $w \leq 15$. Priya's inequality $w \geq 15$ describes weights of 15 pounds and heavier, which is the opposite group. She kept the boundary number but pointed the symbol the wrong way.

2 points	Student explains that 'no more than' sets a maximum the weight cannot pass, so the symbol must be less than or equal to, and writes $w \leq 15$.
1 point	Student writes $w \leq 15$ but does not explain why 'no more than' sets a maximum.
0 points	Student agrees with Priya or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Write two different real-world situations: one that uses $>$ and one that uses $\geq$. Explain why the situations need different symbols even though both mean 'bigger.'

Model answer: The only difference between $>$ and $\geq$ is whether the boundary number itself is allowed. A rule like "score more than 10 points to win" leaves 10 out, so it needs $>$, while "you must be at least 13 to watch" lets exactly 13 count, so it needs $\geq$. Testing the boundary number in your own situation is the quickest way to check which symbol fits.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.